distance, and building type. The output is one of several predefined packages (e.g., *Small Truck + 2 Movers*, *Medium Truck + 3 Movers*, etc.).

- **Random Forest Classifier**:
  - Works by constructing multiple decision trees and averaging their predictions.
  - Robust against overfitting and interpretable due to **feature importance scores**.
  - Well-suited for **tabular datasets** such as customer inventories.

- **Gradient Boosting Machines (GBM)**:
  - Builds models sequentially to correct errors from previous models.
  - High accuracy but more complex to train and interpret.

For **Shifty**, the **Random Forest Classifier** was chosen due to its **balance of performance and interpretability**.

2.3.2 Regression Algorithms for Dynamic Pricing

Predicting moving costs is naturally a **regression problem**, where the goal is to estimate a continuous value (price).

- **Multiple Linear Regression (MLR)**:
  - Establishes a linear relationship between inputs (distance, volume, demand) and price.
  - Simple but struggles with complex, non-linear relationships.
- **XGBoost Regressor**:
  - An advanced implementation of gradient boosting.
  - Handles **feature interactions** effectively and is optimized for both speed and accuracy.
  - Popular in industry and research for predictive modeling.